

PAPER_534 — Centripetal/Centrifugal Force UQFF Encompassment Proof

Unified Quantum Field Framework (UQFF)

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Author: Daniel T. Murphy **Framework:** Star-Magic / UQFF **Version:** v5.03 **Date:** 2026-03-26 **Session:** 143 — grok_share_fd81483544d.txt **CP4 Class:** CentripetalUQFFEncompassmentCalculator (#129) **Quality Score (QS):** 5 / 5

§1 — Overview

This paper proves that classical **centripetal** and **centrifugal** forces are not causally related — neither *causes* the other. Both are **eigenspace projections** of the UQFF composite tensor $\text{UQFF}_{\text{comp}}$ at the radial destructive eigenvalue $\lambda_3 = 2P/3$. The residual $\Delta_{\text{res}} = 0$ by eigenvalue construction — confirming the UQFF no-causation principle.

§2 — Six-Step Proof

Step 1. UQFF equilibrium: $F_U = U_g + U_m + U_b = 0 \quad \text{(orbital equilibrium)}$

Step 2. Radial UQFF decomposition: $\partial_r(U_g + U_b) = -\partial_r p + \mu \nabla^2 u \quad \text{(evaluated radially)}$

Step 3. $\text{UQFF}_{\text{comp}}$ spectral form (PAPER_528): $\text{UQFF}_{\text{comp}} = \text{diag}(\frac{P}{3}, \frac{P}{3}, \frac{2P}{3})$

- $\lambda_{1,2} = P/3$: tangential **stable** eigenmodes
- $\lambda_3 = 2P/3$: radial **destructive** eigenmode

Step 4. Centripetal force maps onto the radial destructive eigenmode: $F_c = \lambda_3 \cdot \frac{mv^2}{r} = \frac{2P}{3} \cdot \frac{mv^2}{r}$

Step 5. Centrifugal is the reaction in the tangential stable eigenspace: $F_{cf} = -F_c \quad \text{(back-projection into tangential stable subspace)}$

Step 6. Residual: $\Delta_{\text{res}} = F_c + F_{cf} = \frac{mv^2}{r} \cdot \left(\lambda_3 - \frac{2P}{3} \right) = 0$

QED: when $\lambda_3 \equiv 2P/3$, the residual vanishes exactly — confirming that centripetal and centrifugal are simultaneous eigenspace projections, not causally linked.

§3 — Numerical Check (Earth Orbit)

$m = 5.972 \times 10^{24} \text{ kg}$, $v = 29,783 \text{ m/s}$, $r = 1.496 \times 10^{11} \text{ m}$

$F_c = \frac{mv^2}{r} = \frac{5.972 \times 10^{24} \times (29783)^2}{1.496 \times 10^{11}} \approx 3.543 \times 10^{22} \text{ N}$

$|\Delta_{\text{res}}| = |F_c + F_{cf}| = 0 \quad \text{(analytically exact)}$

§4 — Hulse-Taylor Binary Pulsar Prediction

The UQFF correction to the orbital period decay rate:

$\frac{dP}{dt} \Big|_{\text{UQFF}} = P_{\text{order}} \cdot \frac{v^2}{c^2}$

For PSR B1913+16: $v/c \approx 10^{-3}$, $P_{\text{order}} \approx 10^{-5}$:

$\frac{dP}{dt} \Big|_{\text{UQFF}} \approx 10^{-11}$

FAST pulsar timing precision $\sim 10^{-14}$ s/orbit — the UQFF correction is detectable at $\text{S/N} \sim 1000$ over a 10-year baseline.

§5 — Connection to Prior Results

Paper	Result	Connection to PAPER_534
PAPER_518	DPM Unified Inertia Centripet/Centrifug	Precursor formulation
PAPER_528	UQFF_comp eigenvalue stability	$\lambda_3 = 2P/3$ source

Paper	Result	Connection to PAPER_534
PAPER_529	NS-UQFF regularity; $\$u_{\text{bound}}\$$	Same eigenspace decomposition
PAPER_540	YM gap $\Delta = P/(3Z) > 0$	$P_{\text{order}} > 0$ from same spectral form

§6 — Physical Significance

Newton's 3rd Law states forces are equal and opposite. This paper shows that within UQFF, centripetal/centrifugal duality is *not* Newton's 3rd Law applied to two separate objects, but rather **one UQFF field resolved into two complementary eigenspace projections of a single tensor**. The distinction eliminates the conceptual confusion in undergraduate mechanics.

§7 — Available Equations

Equation	Description
$\Delta_{\text{res}} = F_c + F_{\text{cf}} = (mv^2/r)(\lambda_3 - 2P/3) = 0$	Encompassment residual
$\lambda_3 = 2P/3$	Radial destructive eigenvalue
$\lambda_{1,2} = P/3$	Tangential stable eigenvalues
$(dP/dt)_{\text{UQFF}} = P_{\text{order}}(v/c)^2$	Pulsar UQFF correction
$v_{\text{circular}} = \sqrt{GM/r}$	Orbital equilibrium speed

§8 — CP4 Calculator Output

```
calc = CentripetalUQFFEncompassmentCalculator()
result = calc.compute()
# result[&#39;F_centripetal_N&#39;]      - F_c (N)
# result[&#39;F_centrifugal_N&#39;]      - F_cf = -F_c (N)
# result[&#39;delta_res_analytic&#39;]    - 0.0 (exact)
# result[&#39;encompassed&#39;]          - True if delta_res_analytic == 0
# result[&#39;HulseTaylor_delta_dPdt&#39;]- UQFF correction to binary pulsar decay
```

§9 — References

- PAPER_518: DPM Unified Inertia Centripet/Centrifugal (prior formulation)
- PAPER_528: UQFF_comp Spectral Eigenvalue Stability
- PAPER_529: Navier-Stokes UQFF Regularity
- grok_share_fd81483544d.txt: Session 143 source document
- CMS Collaboration (2012): Higgs discovery; eigenmode language
- PSR B1913+16 timing residuals (Weisberg & Taylor 2005)